

Justin Richard Burton

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Education:

Brigham Young University, Ira A. Fulton College of Engineering

- B.S. Mechanical Engineering, minor in Mathematics, April 2028, 3.78 GPA

Strengths and Skills:

- **Engineering:** Onshape, Autodesk Inventor, rapid prototyping & additive manufacturing, technical drawings, Arduino, PIC microcontrollers, sensors, motor control, embedded programming, PWM configuration
- **Research:** Technical writing, peer-reviewed publishing, conference presentations (ASME IDETC, UCUR)
- **Programming:** Python (NumPy, SciPy, Matplotlib), C/C++, Java, MATLAB, JavaScript, HTML/CSS, Bash, Git
- **Web & Analytics:** Custom HTML/CSS/JS tooling, JSON catalogs, AI Agents, SEO, Google Analytics
- **Software & Media Tools:** Affinity Designer, Adobe Creative Cloud, Microsoft Office Suite, Google Workspace.
- **Languages:** English (native), Indonesian (fluent), Spanish (foundational)

Work and Volunteer Experience:

Design Exploration Research Group (DesX)

Provo, UT

Research Assistant

August 2025 – Present

- First-authored research on predictive human behavior models in engineering design, presented at ASME IDETC and UCUR
- Managed a research team investigating predictive models of human behavior in the engineering design process
- Contributed to graph-based social impact modeling (GBSIM) tools designed to help engineers assess product impacts

Head of Post-Publication Strategy

June 2026 – Present

- Redesigned and launched the DesX research website, increasing unique visitors 485%, page views 592%, multi-page sessions per user 6.4%, and Google click-through rate 14%
- Developed custom HTML/CSS/JavaScript tooling and GitHub-hosted JSON catalogs for searchable publications, people, tools, and courses, enabling live filtering and content updates without full site rebuilds

HeartRayt (Bench to Bedside 2025 entry)

Provo, UT

Engineering Team Member

September 2024 – April 2025

- Developed a biomedical product including ideation, laboratory validation, establishing design constraints and various stages of design creation and reiteration.
- Evaluated FDA and regulatory considerations and interviewed biomedical professionals to identify and refine product requirements
- Shadowed clinicians in clinical and surgical settings to inform engineering design decisions

Memberships, Awards, & Accomplishments:

- ASME/NSF IDETC Student Essay Competition Winner (2026)
- BYU Biomedical Engineering Association Leadership (2024 - present)
- Engineers in Business Club member (2024 - present)
- Social Media Marketing Certification (2022, renewed in 2024)
- BYU Academic Scholarship (2021-22, 2024 - present)
- Arizona Technical Skills Engineering Certification (2021)
- Rural & Small Town Scholars Award (2021)
- Eagle Scout Award (2018)

Publications:

- Burton, J. R. et al. (2026). "Predictive Models of Human Behavior in Engineering Design." *Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE) 2026*.